

Numerical Analysis

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Topic :- Bairstow Method

Class :- M.Sc. (Mathematics) I Sem I.

Formulae:-

$$\Delta p = - \frac{b_n c_{n-3} - b_{n-1} c_{n-2}}{c_{n-2}^2 - c_{n-3}(c_{n-1} - b_{n-1})}$$

$$\Delta q = - \frac{b_{n-1}(c_{n-1} - b_{n-1}) - b_n c_{n-2}}{c_{n-2}^2 - c_{n-3}(c_{n-1} - b_{n-1})}$$

$$p_1 = p_0 + \Delta p$$

$$q_1 = q_0 + \Delta q$$

For computing b_k 's and c_k 's we use following representation

	a_0	a_1	a_2	...	a_{n-2}	a_{n-1}	a_n
$-p$							
$-q$		$-pb_0$	$-pb_1$	...	$-pb_{n-3}$	$-pb_{n-2}$	$-pb_{n-1}$
			$-qb_0$		$-qb_{n-3}$	$-qb_{n-2}$	$-qb_{n-1}$
$-p$	b_0	b_1	b_2	...	b_{n-2}	b_{n-1}	b_n
$-q$		$-pc_0$	$-pc_1$	...	$-pc_{n-3}$	$-pc_{n-2}$	
			$-qc_0$		$-qc_{n-4}$	$-qc_{n-3}$	
	c_0	c_1	c_2	...	c_{n-2}	c_{n-1}	

Que. Perform two iterations of the Bairstow method to find the quadratic factor of polynomial $x^3 - 2x^2 + x - 2$.

Solⁿ°. Let $x^2 + px + q$ be the quadratic factor.
 Let $\begin{matrix} p_0 = 1 \\ q_0 = 1 \end{matrix}$ } be the initial approximation.

1) First Iteration,

	1	-2	1	-2
$-p_0 = -1$	0	-1	3	-3
$q_0 = -1$	0	0	-1	3
$-p_0 = -1$	$1 = b_0$	$-3 = b_1$	$3 = b_2$	$-2 = b_3$
$-q_0 = -1$	0	-1	4	
	0	0	-1	
	$1 = c_0$	$-4 = c_1$	$6 = c_2$	

$$\Delta p = - \frac{b_3 c_0 - b_2 c_1}{c_1^2 - c_0(c_2 - b_2)} = - \frac{(-2)(1) - (3)(-4)}{(-4)^2 - 1(6 - 3)}$$

$$= - \frac{10}{13}$$

$$\Delta p = -0.7692307692$$

$$\Delta p = -0.769$$

$$\Delta q = - \frac{b_2(c_2 - b_2) - b_3 c_1}{c_1^2 - c_0(c_2 - b_2)} = - \frac{3(6 - 3) - (-2)(-4)}{(-4)^2 - 1(6 - 3)}$$

$$= - \frac{1}{13}$$

$$\Delta q = -0.07692307692$$

$$\Delta q = -0.0769$$

The improved values of p & q are

$$p_1 = p_0 + \Delta p = 1 - 0.769$$

$$p_1 = 0.231$$

$$q_1 = q_0 + \Delta q = 1 - 0.0769$$

$$q_1 = 0.9231$$

2) Second Iteration

	1	-2	1	-2
$-p_1 = -0.231$	0	-0.231	0.5154	-0.1368
$-q_1 = -0.9231$	0	0	-0.9231	2.0594
	$1=b_0$	$-2.231=b_1$	$0.5923=b_2$	$0.0774=b_3$
$-p = -0.231$	0	-0.231	0.5687	
$-q = -0.9231$	0	0	-0.9231	
	$1=c_0$	$-2.462=c_1$	$0.2379=c_2$	

$$\Delta p = \frac{b_3 c_0 - b_2 c_1}{c_1^2 - c_0(c_2 - b_2)} = \frac{0.0774 \times 1 - 0.5923 \times -2.462}{(-2.462)^2 - 1(0.2379 - 0.5923)}$$

$$= -0.2393516114$$

$$\Delta p = -0.2394$$

$$\Delta q = \frac{b_2(c_2 - b_2) - b_3 c_1}{c_1^2 - c_0(c_2 - b_2)} = \frac{0.5923(0.2379 - 0.5923) - (-2.462)(0.0774)}{(-2.462)^2 - 1(0.2379 - 0.5923)}$$

$$= 0.00301633269$$

$$= 0.0030$$

$$p_1 = p_0 + \Delta p = 0.231 - 0.2394 = -0.0084 \sim 0$$

$$q_1 = q_0 + \Delta q = -0.9231 + 0.0030 = -0.9201 \sim 1$$

Hence, the quadratic factor is $x^3 - 2x^2 + x - 2 = x^2 + 1$.